Name: Class: Date:

Activity 1 · Hands-on · 🏠 At home · ⏱ 5 days · 10 min a day

🫙 A Pond in a Jar (Hay Infusion)

You will learn: Dormant microbes on dry grass come back to life — first bacteria (Monera), then protozoans (Protista) that eat them.

YOU NEED

- A clean glass jar or bottle
- A handful of dry grass or hay
- Tap water left open for one day (so chlorine escapes)
- A thin cloth and a rubber band
- A dropper or straw, a glass slide or clear plastic sheet
- Phone microscope (BAIO) or a hand lens

⚠️ **Safety:** Wash your hands after every check. Do not drink or smell the water closely. Pour it down the drain at the end.

STEPS

1. Put the dry grass in the jar and fill it three-quarters with the water.
2. Cover with the cloth (not a lid — air must enter) and keep it in a warm place away from direct sun.
3. Every day, note the colour, smell and any film on the surface.
4. From day 3, take one drop from the surface film and place it on the slide.
5. Observe with the phone microscope at the highest zoom. Look for moving organisms and sketch them.
6. Compare day 3 with day 5 — which organisms became more common?

OBSERVATION TABLE

Day	Colour & smell	Surface film?	Moving organisms seen

QUESTIONS

1. Why did the water turn cloudy before you could see Paramecium?

2. To which kingdom does Paramecium belong? How does it move?

3. Why was the jar covered with cloth and not a lid?

WHAT IS HAPPENING? — THE SCIENCE

Dry grass carries bacterial spores and protozoan cysts. In water they become active. Bacteria multiply first and make the water cloudy. Protozoans such as Paramecium (ciliated) then appear and feed on the bacteria — showing kingdom Monera and kingdom Protista living side by side.

Key terms: Monera · Protista · cyst · ciliated protozoan

Activity 2 · Hands-on · 🏠 At home · ⏱ 7 days · 5 min a day

🍞 Grow a Mould Garden

> **You will learn:** Fungi grow from spores in the air and form hyphae and a mycelium — they are saprophytic heterotrophs.

YOU NEED

- 2 slices of bread (or a piece of roti) and an orange peel
- 2 zip-lock bags or plastic boxes with lids
- A few drops of water
- Marker pen and tape

> ⚠️ **Safety:** Never open the bags once sealed — mould spores can cause allergies. If you have asthma or allergies, only observe. Throw the bags away sealed.

STEPS

1. Wipe the bread and peel across a table or floor to collect spores.
2. Sprinkle a few drops of water on one slice; keep the other slice dry.
3. Seal each item in its own bag and label: "moist", "dry", "peel".
4. Keep the bags in a dark, warm place.
5. Each day, look through the bag and note the colour and area of mould.
6. On day 7, use the phone microscope through the bag to see the fuzzy threads.

OBSERVATION TABLE

Day	Moist bread	Dry bread	Orange peel

QUESTIONS

1. Which bag grew mould fastest? Why?

2. Why are fungi kept in a separate kingdom and not with plants?

3. Name the fungus that gave the first antibiotic.

WHAT IS HAPPENING? — THE SCIENCE

Fungal spores are everywhere in air. On moist food they germinate into thread-like hyphae; the network of hyphae is the mycelium. Black mould is often Rhizopus; blue-green mould is often Penicillium — the fungus that gave us penicillin. Fungi cannot make food, so they absorb it from dead matter (saprophytes).

Key terms: **hyphae** **mycelium** **saprophyte** **Rhizopus** **Penicillium**

Digital activity

💻 Kingdom Sorting Desk

> Tap an organism, then tap the kingdom it belongs to.

Do this digital activity in the BAIO app: **https://narendra-bio-sir.github.io/BAIO-Learning-System-By-Narendra-Sir/activities.html#pu1-ch2/2**

MONERA · PROTISTA · FUNGI · PLANTAE · ANIMALIA

Monera	Protista	Fungi	Plantae	Animalia

Nostoc · Mycoplasma · Methanogen · Euglena · Paramecium · Diatom · Plasmodium · Mushroom (Agaricus) · Yeast · Rhizopus · Puccinia · Moss (Funaria) · Mango tree · Fern · Earthworm · Starfish · Frog

> **You will learn:** Whittaker's five kingdoms separate prokaryotes (Monera), single-celled eukaryotes (Protista), absorbers (Fungi), photosynthesisers (Plantae) and ingesters (Animalia).

THINK & WRITE

1. Why is Euglena hard to place — plant or animal?

2. Why is yeast placed in Fungi even though it is single-celled?